📊 US Pharma Sales Analytics Pipeline – HER2+ Breast Cancer

1. Objective & Business Context

- **Goal:** Track **Kadcyla** and **Trastuzumab** performance vs competitors (**Perjeta, Enhertu, Phesgo**) in the **HER2+ breast cancer market** in the US.
- **Key Questions:**
  1. Market share by state
  2. MoM / YoY growth trends
  3. Drug rankings in priority states (CA, NY, TX, FL)
  4. Rolling trends (R3M, R6M)
  5. Top prescribers contributing to growth
- **Approach:**
  1. Raw data → Cleaning → Aggregation → Business rules → Reporting tables → Dashboards
  2. Pipeline runs **weekly** (~6–7 hours via Azkaban)

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2. Architecture

S3 → ETL → Redshift → Azkaban (6–7h weekly) → Reporting Tables → Power BI Dashboards

- **S3:** Raw prescription data
- **ETL:** Transform & load into Redshift
- **Azkaban:** Orchestration & scheduling
- **Redshift:** Multi-layer warehouse
- **Power BI:** Executive dashboards

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3. Pipeline Layers & Tables

Layer	Purpose	Tables (3–5 per layer)
LDG	Raw capture	raw_sales, daily_agg, drug_profile, s3_qc, volume_summary
STG	Cleaning & QC	clean_sales, qc_nulls, qc_duplicates, qc_negatives, region_mdm

Layer	Purpose	Tables (3–5 per layer)
FACT	Aggregation	daily_fact, monthly_fact, provider_fact, drug_fact, state_fact
INT	Rolling metrics	rolling_r3m, rolling_r6m, ytd_metrics, trend_analysis
MDM	Master data	drug_mdm, region_mdm, provider_mdm, competitor_mdm
BUS	Business rules	oncology_sales, portfolio_classification, monthly_classified, state_filter
RPT	Reporting/KPIs	kpi_monthly, us_summary, competitor_benchmark, top_providers, state_summary

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4. Key KPIs

KPI	Formula
TRx	SUM(trx)
NRx	SUM(nrx)
CbNrx	SUM(cbnrx)
Total Sales	SUM(sales_value)
R3M Sales	AVG(total_sales) OVER 3 months
R6M Sales	AVG(total_sales) OVER 6 months
YTD Sales	SUM(total_sales) OVER (YEAR)
Market Share	total_sales / SUM(total_sales) OVER (region, month)
Rank	RANK() OVER (region, month ORDER BY total_sales DESC)
MoM Growth	(total_sales - prev_month_sales) / prev_month_sales

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5. Dashboard Design

- **Line Charts:** R3M/R6M trends by drug & region
- **Bar Charts:** Market share comparison

- **Maps:** US states with total sales bubbles
- **Tables:** Top providers by NRx growth
- **KPI Cards:** TRx, NRx, CbNrx, Total Sales, R3M, R6M, YTD, Market Share, Rank, MoM Growth

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